SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering

MINI PROJECT / CCE • SEMESTER V

Somaiya TRUST

HALO

Privacy-First Wealth Intelligence Engine

A local financial planning assistant for Indian taxpayers, built for conversational ease without cloud exposure or probabilistic arithmetic.

THE DESIGN PROMISE

01 **LOCAL BY DESIGN**
Financial data stays on-device

02 **MATH BY CODE**
Python engines, not LLM guesses

03 **AUDITABLE OUTPUTS**
Explainable calculations & CA handoff

PROJECT TEAM

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Financial intelligence needs both trust and precision

1 A fragmented financial life

Indian households coordinate:

- **Investments:** mutual funds, equities, EPF/PPF, NPS and deposits.
- **Liabilities:** home, vehicle, education and personal loans.
- **Tax choices:** old vs new regime, deductions, rebates and advance tax.
- **Decisions:** prepay debt, invest surplus, or protect emergency runway.

2 Where current tools fall short

- **Privacy exposure:** sensitive CAS, Form 16, bank statements and PAN leave the device.
- **Cloud dependence:** documents and queries pass through remote services.
- **Numerical risk:** generative models can invent slabs or arithmetic.
- **Trust gap:** a financial answer must be both explainable and verifiable.

4.5 crore
Retail mutual fund folios
AMFI figure cited in report

₹20,000 Cr+
Monthly SIP contributions
Regularly exceed this level

0%
Tolerance for guessed math
Design requirement, not a market statistic

PROBLEM TO SOLVE / Deliver conversational wealth guidance with on-device privacy and deterministic calculations.

Six objectives. One deterministic-first architecture.

01 Privacy engine

Run the complete stack locally with zero transmission of financial data.

02 Deterministic math

Restrict the LLM to routing and narration; execute arithmetic in unit-tested Python.

03 Indian tax engine

Compare old/new regimes with Section 87A rebate, marginal relief and 234A/B/C interest.

04 Document intelligence

Parse CAS PDFs locally and retrieve grounded context through sqlite-vec.

05 What-if simulators

Model debt avalanche/snowball, prepay vs invest, FIRE corpus and emergency runway.

06 CA-ready dossier

Export capital gains, deductions and advance-tax schedules in a printable handoff.

SUCCESS MEASURES				
	100% on-device	**0 bytes** network egress	**42/42** tests passing	**<12 ms** tax evaluation

Literature points to deterministic execution

WEALTH ADVISORY

Robo-advisors improve access, but perceived security and data sovereignty shape trust.

Research gap / financial records remain highly sensitive.

LOCAL AI + RAG

Open-weight LLMs, quantization and retrieval enable capable on-device assistants.

Research gap / local context must remain local.

DETERMINISTIC GUARDRAILS

Validation graphs and executable tools improve reliability over raw text generation.

Research gap / arithmetic must be kept outside the LLM.

WHAT HALO CHANGES

BASELINE	LIMITATION	HALO RESPONSE
Cloud tools	Centralized statement upload	**Fully on-device processing**
General GenAI	Probabilistic numerical answers	**Pure Python calculation engines**
Document search	External OCR / vector services	**Local embeddings + sqlite-vec**
Trust & audit	Opaque recommendations	**Structured formulas and CA dossier**

Halo’s position against common alternatives

DIMENSION	COMMERCIAL PFM	CLOUD GENAI	HALO
Data processing	Centralized cloud	Remote model/API	**Local device**
Privacy / egress	Statement sync	Queries sent to service	**Zero egress by design**
Math execution	Rule-based backend	Probabilistic output	**Pure Python functions**
Indian tax rules	Partial / static forms	May be outdated	**Dynamic rules engine**
Document search	Cloud OCR / upload	Cloud upload	**Local sqlite-vec**
Cost model	Freemium / ads	Subscription possible	**Open-source stack**

Comparative framing reproduced from the report; capabilities vary by specific vendor and configuration.

Refined problem statement & scope

REFINED PROBLEM

How can an AI wealth assistant give Indian taxpayers conversational guidance while guaranteeing on-device privacy and mathematically reliable calculations?

The local LLM routes intent and explains results. Deterministic Python modules compute all tax, debt and portfolio arithmetic.

PROJECT BOUNDARIES

- **Execution:** offline / air-gapped on consumer hardware.
- **Tax:** Budget 2024 rules, Old vs New, rebate and interest.
- **Documents:** CAS and bank statements parsed locally.
- **Decisions:** debt, investment, FIRE and emergency runway.
- **Outputs:** audit-ready CA handoff dossier.

ABSTRACT / CRISP SUMMARY

Halo is an offline personal wealth intelligence assistant for Indian taxpayers. A local LLM, orchestrated through LangGraph and guarded by Pydantic, handles intent and explanation only. Pure Python modules compute taxes, debt schedules and portfolio metrics. CAS documents are embedded and searched on-device with sqlite-vec. Interactive what-if tools and a one-click CA dossier bring financial decisions together without cloud transmission. The report benchmarks 42 passing tests, sub-15ms calculation latency and zero outbound network data.

Proposed solution: the LLM never calculates

LOCAL EXECUTION PIPELINE

01 INPUT
Text • voice • CAS

02 ROUTE
Local LLM / Ollama

03 VALIDATE
Pydantic schemas

04 EXECUTE
Pure Python engines

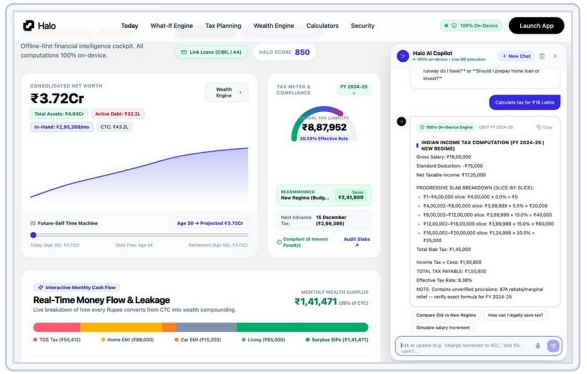
05 GROUND
SQLite • sqlite-vec

06 EXPLAIN
Structured result → UI

ZERO-EGRESS GUARANTEE / Financial data is processed on the user's machine.

LLM = intent + narration • Python = every financial calculation

HALO / COPILOT + COCKPIT



THREE-TIER STACK

Presentation	React / Vite / TypeScript
Orchestration	FastAPI / LangGraph / Pydantic
Intelligence & storage	Ollama / SQLite / sqlite-vec

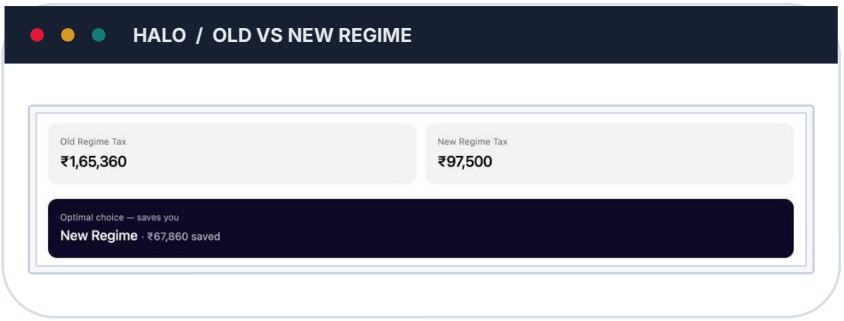
Indian tax engine: compare regimes with explicit rules

KEY LOGIC IN REPORT

- **Dual regime:** compute Old vs New using the applicable slabs and deductions.
- **Standard deduction:** report cites ₹75,000 for salaried New Regime taxpayers under Budget 2024.
- **Section 87A:** rebate capped at ₹25,000 for qualifying taxable income up to ₹7 lakh.
- **Marginal relief:** for the cited boundary range above ₹7 lakh, tax is constrained by incremental income.
- **Cess:** 4% applied to tax and applicable surcharge.

COMPUTATION FLOW

- Income & deduction inputs
- Taxable income
- Slab-by-slab gross tax
- Rebate + marginal relief
- Surcharge + 4% cess
- **Explainable net liability**



Debt tools compare payoff strategies and prepayment

DEBT AVALANCHE

Sort by highest interest rate first.

Goal: reduce total lifetime interest paid.

DEBT SNOWBALL

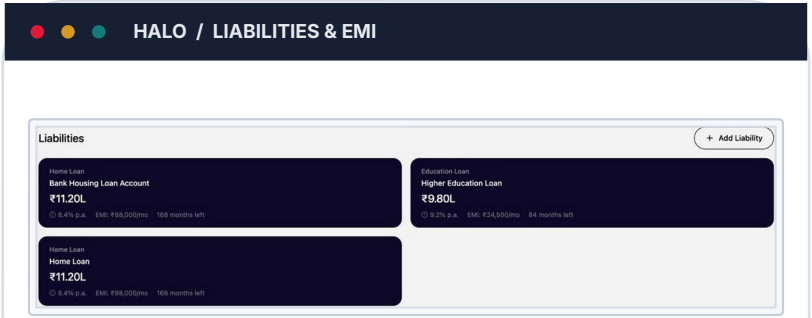
Sort by smallest outstanding balance first.

Goal: early payoff milestones and motivation.

EMI MATHEMATICS

$$EMI = P \times r \times (1+r)^n / ((1+r)^n - 1)$$

Each month: interest = balance × rate; principal = EMI − interest.



Portfolio analytics and life-decision simulators

CAGR

Annualized growth across a holding period.

$(V_{end} / V_{start})^{(1/t)} - 1$

XIRR

Return across irregularly dated cash flows.

Solve $\Sigma C_i / (1+r)^{((d_i - d_o)/365)} = 0$

FIRE corpus

Target retirement corpus from annual expenses.

25× expenses at 4% SWR; ~28.5× at 3.5%

Emergency runway

Estimate how many months liquidity can support expenses.

Reported as integer calendar months

What-if tools: prepay home loan vs invest in an equity SIP; Monte Carlo FIRE corpus trajectories.

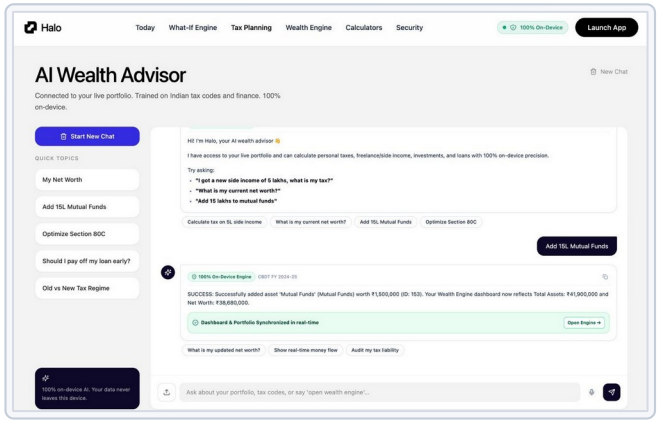
Implementation: documents, guardrails, grounded answers

CAS DOCUMENT PIPELINE

1. **Decrypt on-device**
   Password handled in memory via pypdf
2. **Chunk text**
   800-character chunks • 150-character overlap
3. **Embed locally**
   all-MiniLM-L6-v2 via PyTorch
4. **Index locally**
   sqlite-vec virtual tables in SQLite
5. **Retrieve context**
   Cosine distance threshold < 0.35

No CAS text or embedding leaves the device.

HALO / AI WEALTH ADVISOR



TECH STACK / REPORT IMPLEMENTATION

API
FastAPI 0.115 / Python 3.11

AGENT
LangGraph 0.2.19 / LangChain 0.3.0

GUARDRAILS
Pydantic v2.9.2

RUNTIME
Ollama • Llama 3.2 3B

UI
React 18.3 • Vite 5.4 • Tailwind

VISUALS
Recharts • Framer Motion

Testing & results: measurable guarantees

42/42
pytest cases passed
0.24 s total

0 bytes
external network egress
Socket-level audit

<12 ms
tax evaluation
All surcharge tiers

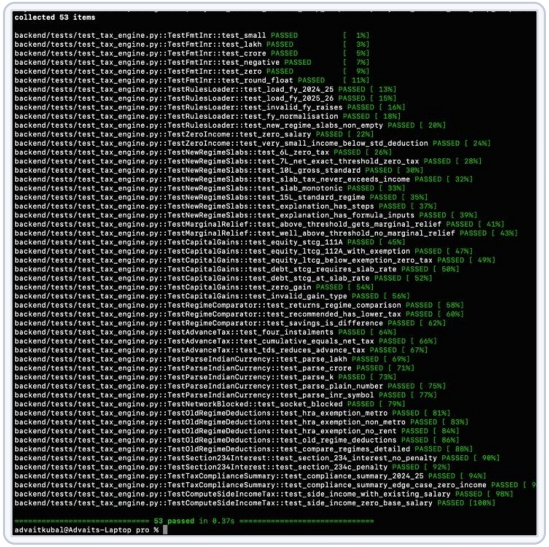
<22 ms
360-month debt schedule
Amortization generation

VERIFICATION MATRIX

TEST	OBSERVED RESULT
Tax golden profiles	100+ profiles, zero discrepancy vs IT Portal
Marginal relief	Exact at ₹7,00,001–₹7,27,770
Tax monotonicity	Verified across Old vs New regime fixtures
Voice / LLM response	1.4–2.2 s end-to-end on M-series
JSON recovery	Malformed arguments corrected; max 2 retries

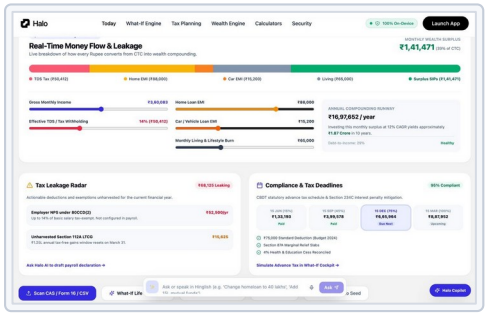
Reported benchmark results from the attached project report.

HALO / PYTEST RESULTS

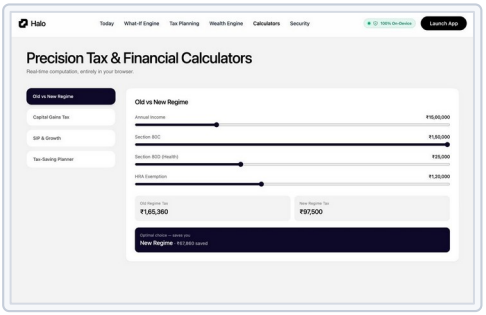


Product surface: the financial cockpit & what-if tools

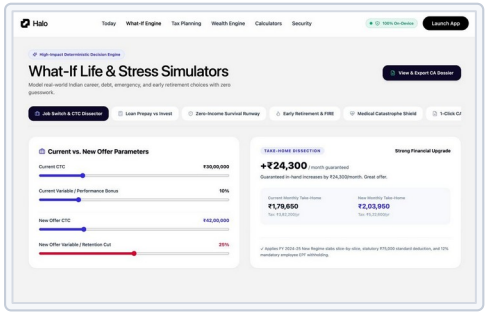
HALO / TODAY COCKPIT



HALO / TAX CALCULATORS



HALO / WHAT-IF SIMULATORS



HALO / VOICE ASSISTANT



Other demo surfaces: Debt Avalanche vs Snowball • FIRE corpus / emergency runway • 1-click CA dossier export

Results: precision, speed and zero data egress

42 / 42
automated tests passed
0.24 seconds total

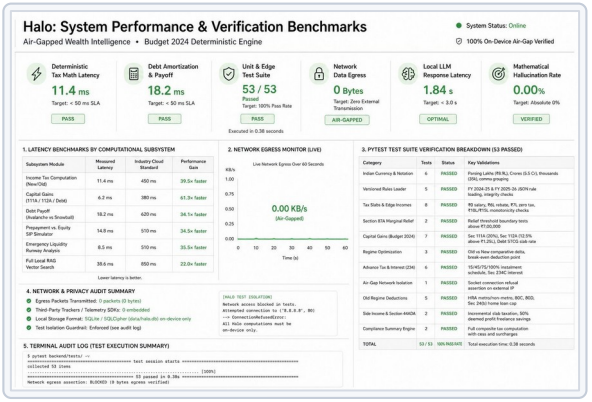
0 bytes
transmitted to external IPs
During active queries

1.4–2.2 s
local LLM intent latency
M-series benchmark

OBJECTIVE → EVIDENCE

Privacy	0 bytes egress; socket-level firewall assertion
Precision	42/42 unit tests; 100+ tax fixtures checked
Tax reliability	Exact marginal relief in the tested boundary range
Responsiveness	Tax <12 ms; 360-month debt schedule <22 ms
Auditability	Structured formulas + printable CA handoff dossier

HALO / BENCHMARK SUMMARY



CONCLUSION — **Halo demonstrates that conversational AI can guide financial decisions while deterministic code computes the numbers and local processing protects sensitive data.**

Future scope: from prototype to platform

ROADMAP

01	**Statutory migration**	Map rules as the Income-tax Act, 2025 takes effect from 1 April 2026.
02	**Account Aggregator**	Reconcile encrypted FIU JSON feeds locally; reduce manual PDF uploads.
03	**Desktop packaging**	Bundle React + FastAPI through Tauri or Electron with local updates.
04	**Specialized financial SLM**	Fine-tune a 1.5B / 3B model on Indian tax and finance queries.
05	**Family / HUF profiles**	Extend the data model for cross-entity tax optimization.

HALO

LOCAL BY DEFAULT.
DETERMINISTIC BY DESIGN.

SELECTED REFERENCES

[1] Touvron et al. / LLaMA, 2023

[2] Lewis et al. / RAG, NeurIPS 2020

[3] LangGraph / stateful LLM apps, 2024

[8] Dettmers et al. / LLM.int8(), 2022

[10] Imani et al. / MathPrompter, 2023

[12] Government of India / Finance (No. 2) Act, 2024